Experiment-3(tower of hanoi)

Aim:

To solve the Tower of Hanoi problem using recursion.

Algorithm:

1. If there is only one disk, move it directly from the source peg to the destination peg.
2. Otherwise:

   Move n−1 disks from the source peg to the auxiliary peg.

   Move the largest disk (nth disk) from the source peg to the destination peg.

   Move the n−1 disks from the auxiliary peg to the destination peg.

3. Repeat recursively until all disks are moved to the destination peg.

Python Program:

```
python
def tower_of_hanoi(n, source, auxiliary, destination):
    if n == 1:
        print("Move disk 1 from", source, "to", destination)
        return

    tower_of_hanoi(n - 1, source, destination, auxiliary)

    print("Move disk", n, "from", source, "to", destination)

    tower_of_hanoi(n - 1, auxiliary, source, destination)


n = int(input("Enter number of disks: "))
tower_of_hanoi(n, 'A', 'B', 'C')
```

Sample Output (for 3 disks):

```
Enter number of disks: 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
```

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Move disk 1 from A to C